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Digital design and performance evaluation of porous metal-bonded grinding wheels based on minimal surface and 3D printing

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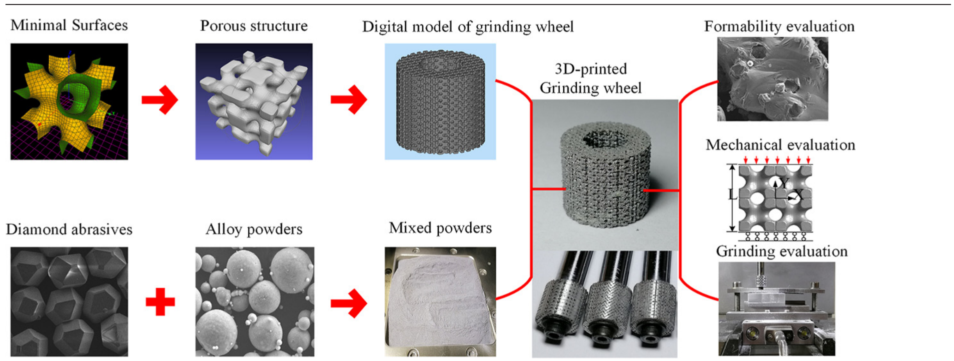
HIGHLIGHTS

- A novel method is proposed to design and fabricate porous grinding wheels based on minimal surfaces and 3D printing.
- The pore morphology and mechanical properties of wheel porous structure can be designed and controlled by this novel method.
- The grinding wheel with well-formed porous structure and firmly-bonded abrasives can be fabricated by this novel method.
- The excellent grinding performance of 3D-printed grinding wheel proves the effectiveness and advantages of this novel method.

GRAPHICAL ABSTRACT



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ABSTRACT

Porous metal-bonded grinding wheel is in great demand but hard to realize with traditional fabrication process. In this article, a novel design and fabrication method of porous metal-bonded grinding wheel is proposed based on minimal surface and 3D printing. With this novel method, numerous interconnected pores and liquid channels can be realized for metal-bonded grinding wheel. Schwarz P, Schwarz D and Schoen I-WP minimal surfaces are adopted to construct the porous structures of grinding wheel. The formability is investigated for 3D-printed binder-abrasive composite and 3D-printed wheel porous structure. The mechanical properties of wheel porous structure are evaluated based on experiment and finite element method. The grinding performance of 3D-printed grinding wheel is evaluated and compared to the electroplated wheel. Results indicate that the pore morphology and mechanical performance of wheel porous structure can be controlled by design parameters. Moreover, grinding wheels with well-bonded abrasives and well-formed pores can be realized by 3D printing. Finally, the excellent grinding performance of 3D-printed grinding wheel proves the effectiveness and advantages of proposed novel design and fabrication method.

1. Introduction

Grinding wheel is extensively applied to the precision machining of key components owing to its excellent grinding performance [1]. The most popular and mature technological process for grinding wheel is hot-pressing sintering, which includes material selection, powder

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mixing, hot pressing and dressing. Compared with resin and ceramic binder, metal binder can contribute to a better performance in terms of abrasive holding ability and anti-deforming ability [2,3]. However, limitations become clear that the porosity is small and inner porous structure cannot be designed and controlled based on traditional hot-pressing sintering technique [4]. Actually, pores, liquid channels and porosity play a significant role in the grinding performance. Lack of pores and liquid channels may lead to small debris space, high working temperature and even surface burn [5–7].

As 3D printing technology becomes more and more mature, its capacity of forming metal matrix composite and sophisticated structure proves to be strong and powerful [8–11]. Thus, 3D printing technology is chosen as the forming process for metal-bonded grinding wheel with pre-designed porous structure (composed of numerous pores and liquid channels). However, the 3D printing of this special composite (composed of metal binder and abrasives) faces big challenge and has been rarely studied before.

Porous structure can help to promote the heat transfer performance and anti-clogging performance of grinding wheel, whose mechanical performance can be controlled by pore morphology and porosity. Lots of methods are utilized to generate porous structures, including Boolean operation method [12], lattice-based method [13], 3D reconstruction method [14] and minimal surface method [15]. In this paper, triply periodic minimal surface (TPMS) is adopted to construct the wheel porous structure. The mean curvature of TPMS is zero everywhere, which indicates that the transition of the minimal surface is smooth and the stress distribution under loading condition is uniform, which can avoid the stress concentration phenomenon. Furthermore, lots of interconnected pores and liquid channels can be constructed by TPMS, which is good for coolant flow and heat dissipation. Besides, TPMS is modeled by mathematical implicit functions, and wheel porous structures can be accurately expressed and regulated by adjusting function parameters. Therefore, TPMS method can realize the digital design of wheel porous structure.

In this study, we propose one novel method to design and fabricate porous metal-bonded grinding wheel based on TPMS and 3D printing. With this method, three TPMS structures, including Schwarz P structure, Schwarz D structure and Schoen I-WP structure, are designed and fabricated with different porosities. Firstly, the design, regulation and fabrication method of wheel porous structure is discussed. Secondly, the formability evaluation of binder-abrasive composite and porous structure is investigated. Thirdly, the mechanical properties of 3D-printed porous structure are evaluated by experimental and finite element method. Finally, the grinding performance of 3D-printed grinding wheel is evaluated.

2. Design method of porous structure

TPMS modeling method enables us to control the porosity, pore size, distribution and morphology by adjusting the design parameters in the implicit function [15]. TPMS are minimal surface model which can be infinitely extended in three periodic directions. Using this surface modeling method, digital porous metal-bonded grinding wheel can be realized. TPMS can be defined by Weierstrass formula which is called parametric equation and shown as follows:

$$\begin{cases} x = Re\int_{\omega_0}^{\omega_1} e^{i\theta}(1-\omega^2)R(\omega)d\omega \\ y = Re\int_{\omega_0}^{\omega_1} e^{i\theta}(1+\omega^2)R(\omega)d\omega \\ z = Re\int_{\omega_0}^{\omega_1} e^{i\theta}(2\omega)R(\omega)d\omega \end{cases} \tag{1}$$

where ω represents a complex variable, θ is Bonnet angle, $R(\omega)$ represents the surface function. Furthermore, the implicit function form of TPMS can be established as follows:

$$\varnothing(r) = \sum_{k=1}^{K} A_k \cos\left[\frac{2\pi(h_k r)}{\lambda_k} + p_k\right] = C \tag{2}$$

where r is the position vectors of Euclidean space, A_k is the amplitude factor, h_k is the k^{th} grid vector in the reciprocal space, λ_k is the periodic wavelength, p_k is the phase offset and C is the offset value of implicit function.

The implicit function form of TPMS is widely used because it is easy to express and calculate even though it is approximate. In this study, three kinds of TPMS surfaces are adopted, including Schwarz P surface, Schwarz D surface as well as Schoen I-WP surface, whose implicit functions are listed in Table 1. These surfaces are 3D constructed and modeled by K3DSurf software, which are depicted in Fig. 1a b and c. This software can be used to generate 3D surface in mesh file type (*.obj) depending on the given implicit function and definition domain. The porous structure of grinding wheel can be generated by Boolean operation between TPMS surfaces and hexahedral boundary, which are demonstrated in Fig. 1d e and f.

To realize the digital design of wheel porous structure, the control equations of porosity P, relative density ρ_R, equivalent aperture d and specific surface area S_v are obtained by least squares fitting, and shown as follows:

$$\begin{cases} P_P = 0.2855C_P + 0.5009 \\ P_D = 0.5834C_D + 0.5031 \\ P_I = 0.1347C_I + 0.4806 \end{cases} \tag{3}$$

$$\begin{cases} \rho_{RP} = -0.2855C_P + 0.4991 \\ \rho_{RD} = -0.5834C_D + 0.4969 \\ \rho_{RI} = -0.1347C_I + 0.5194 \end{cases} \tag{4}$$

$$\begin{cases} d_P = \sqrt[3]{\dfrac{6(0.2855C_P + 0.5009)}{\pi}} \\ d_D = \sqrt[3]{\dfrac{6(0.5834C_D + 0.5031)}{\pi}} \\ d_I = \sqrt[3]{\dfrac{6(0.1347C_I + 0.4806)}{\pi}} \end{cases} \tag{5}$$

$$\begin{cases} S_{vP} = 0.5575C_P^2 + 1.3845C_P + 6.8502 \\ S_{vD} = 8.4661C_D^2 + 8.873C_D + 12.859 \\ S_{vI} = 0.4886C_I^2 + 2.0908C_I + 10.504 \end{cases} \tag{6}$$

where P_P, P_D and P_I represent the porosity of Schwarz P, Schwarz D and Schoen I-WP structures, ρ_{RP}, ρ_{RD} and ρ_{RI} represent the corresponding relative density, d_P, d_D and d_I refer to the corresponding equivalent aperture, S_{vP}, S_{vD} and S_{vI} refer to the corresponding specific surface area, C_P, C_D and C_I are the corresponding offset values of implicit functions.

It reveals that the porosity P, relative density ρ_R, equivalent aperture d and specific surface area S_v can be accurately expressed as the functions of offset value C. Based on the control equations, the geometric feature of wheel porous structure can be designed and regulated by

Table 1
Implicit functions of three kinds of TPMS surfaces used in this study.

TPMS	Periodic implicit function	Definite domain
Schwarz P	$\varnothing(r) = \cos X + \cos Y + \cos Z = C$	$X, Y, Z \in [-\pi,\pi]$
Schwarz D	$\varnothing(r) = \cos X \cos Y \cos Z - \sin(X) \sin(Y) \sin(Z) = C$	$X, Y, Z \in [-\pi,\pi]$
Schoen I-WP	$\varnothing(r) = 2(\cos X \cos Y + \cos Y \cos Z + \cos Z \cos X) - (\cos 2X + \cos 2Y + \cos 2Z) = C$	$X, Y, Z \in [-\pi,\pi]$

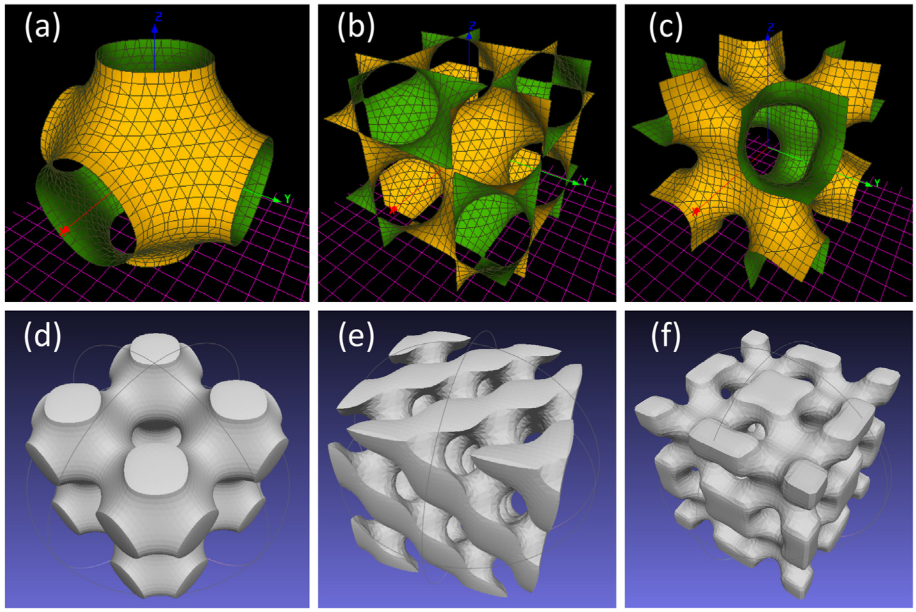

Fig. 1. Porous structure design based on TPMS: (a) Schwarz P surface; (b) Schwarz D surface; (c) Schoen I-WP surface; (d) Schwarz P structure; (e) Schwarz D structure; (f) Schoen I-WP structure.

Table 2
Parameters of metal binder and abrasives used for 3D-printed grinding wheel.

Parameter	Binder	Abrasives
Material	AlSi10Mg alloy	Diamond
Density	2.68 g/cm^3	3.52 g/cm^3
Particle size	15-53 μm	62-75 μm
Particle shape	Sphere	Truncated octahedron
Volume fraction	85%	15%

changing the offset value C, therefore the pore morphology of wheel porous structure can be actively controlled.

To evaluate the influence of porosity on the mechanical properties of TPMS structures, porous structures with porosity of 0.3, 0.4, 0.5 and 0.6 are designed based on Schwarz P, Schwarz D and Schoen I-WP surfaces respectively.

3. Fabrication method of grinding wheel

The raw materials for 3D-printed grinding wheel are composed of metal powders and abrasives, whose parameters are shown in Table 2. The binder powders and abrasives are evenly mixed by the mixing machine for 1 h and dried at 80°C by the vacuum oven for 4 h to obtain the uniform and dry mixed powders. Selective laser melting (SLM) is a powerful metal 3D printing technology and used for the fabrication of 3D-printed grinding wheel in this study. The corresponding fabrication principle is depicted in Fig. 2, and the detailed procedure refers to our previous work [1]. Because of the sophisticated structure, the 3D printing time is about 30 min for a single grinding wheel. The alloy powders are melted as binder around diamond abrasives. The porous 3D-printed grinding wheel will be fabricated layer by layer. Fig. 3 shows the digital models and 3D-printed samples of grinding wheels with pre-designed

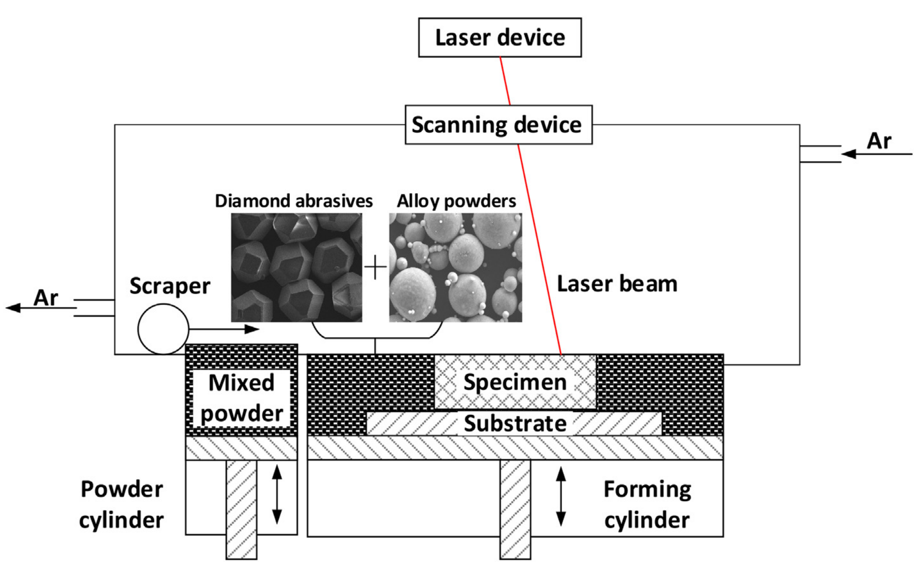

Fig. 2. Fabrication principle of 3D-printed metal-bonded diamond grinding wheel.

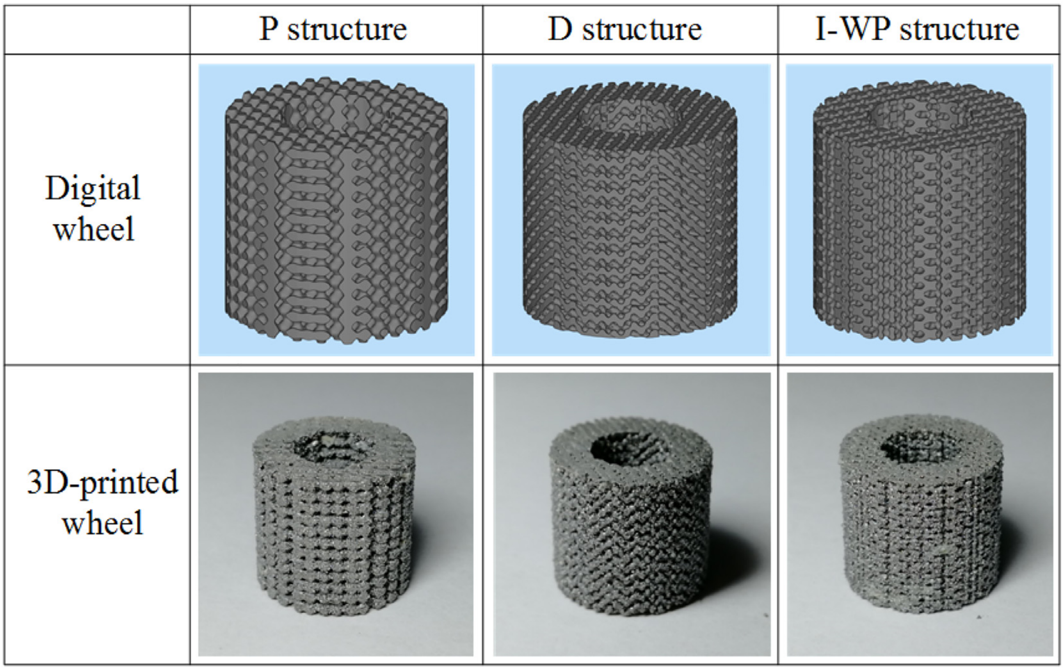

Fig. 3. Digital models and 3D-printed samples of grinding wheels with pre-designed porous structures.

porous structures, including Schwarz P, Schwarz D and Schoen I-WP structures.

4. Formability evaluation of grinding wheel

In the process of laser sintering, the mixed powder is transformed into composite material. To assess the formability of 3D-printed composite, surface morphology of as-built and polished composite is observed using Scanning Electron Microscope (SEM). Meanwhile, to assess the formability of 3D-printed porous structure, macro and micro morphology of Schwarz P structure, Schwarz D structure and Schoen I-WP structure are investigated.

Fig. 4a, b, c and d show the surface morphology of as-built composite with different magnification respectively. The diamond abrasives are embedded into metal matrix with intact polyhedron shape. During SLM process, the metal powder surrounding diamond abrasives are melted and become liquid metal, which will solidify as matrix packing abrasives in it. To evaluate the bonding condition of abrasives, the 3D-printed composite is polished with diamond sandpaper. When the surface become smoothing after times of polishing, the polished surface of composite is observed using SEM, shown in Fig. 4e and f. It can be

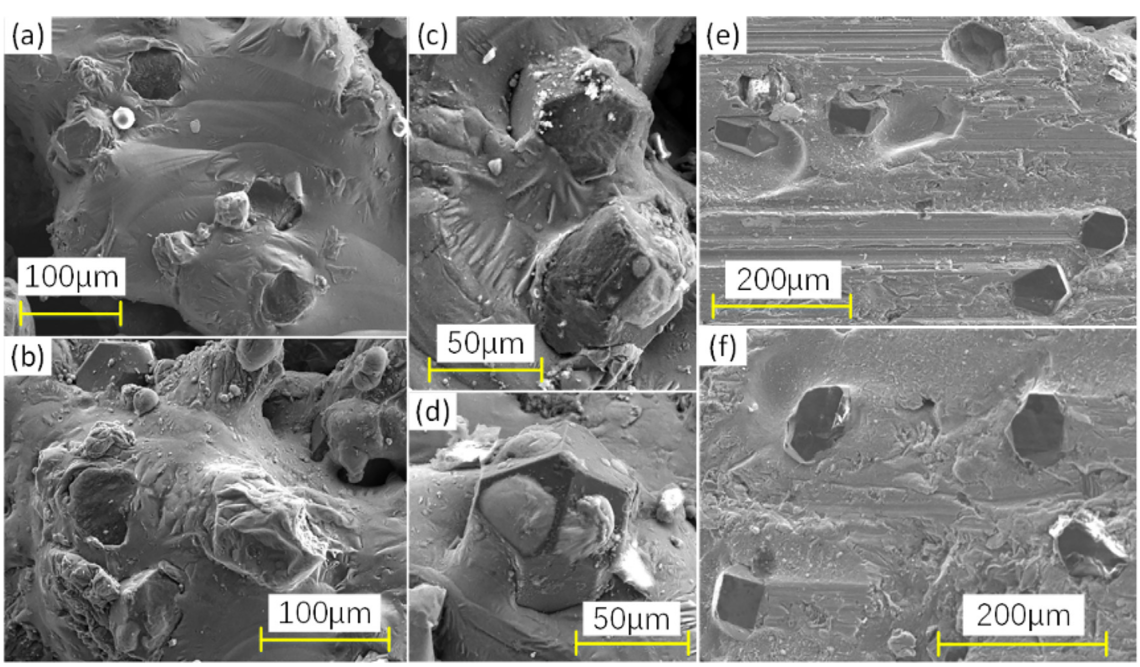

Fig. 4. Morphological analysis of 3D-printed composite: (a-d) surface morphology of as-built composite; (e-f) surface morphology of polished composite.

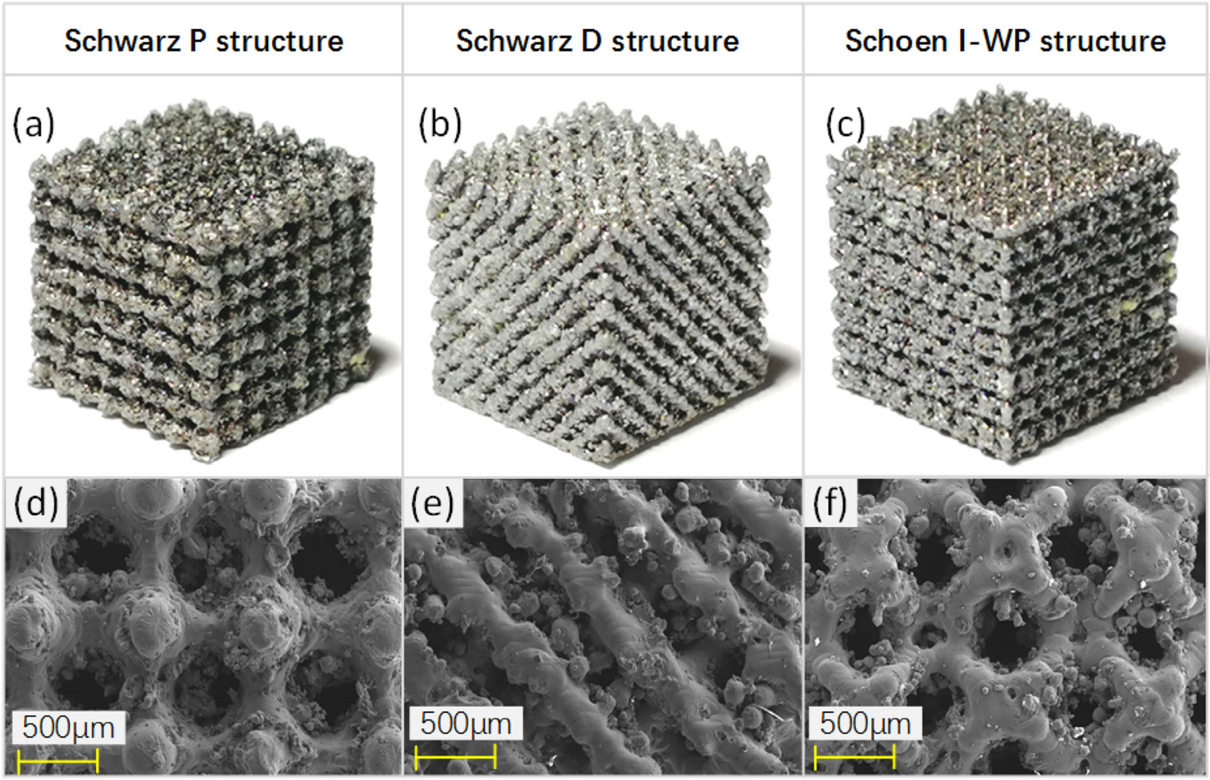

Fig. 5. Morphology of 3D-printed porous structures: (a-c) macro morphology; (d-f) micro morphology.

seen that the bonding interface is reliable and bonding strength is enough because most of abrasives do not drop off during polishing process. The macro morphology and micro morphology of 3D-printed porous structures with porosity of 0.6 are shown in Fig. 5a-c and d-f respectively. It is revealed that, the porous structure is well-formed and every design unit appears clearly visible, indicating the good formability of pre-designed porous structure.

Due to the fluidity of molten metal and forming limits of SLM process, the forming accuracy of porous structure is facing challenging. Because of the difference between design model and formed samples, some errors exist for theoretical and practical porosity of 3D-printed porous structure. As a result, practical porosity is assessed adopting precision balance method and true densitometer method, and compared to corresponding theoretical values. Table 3 shows the measurement results, including mean value, standard deviation and relative error.

Results indicate that the practical porosity is lower than theoretical porosity for all the 3D-printed structures and the relative error becomes bigger as porosity increases. This phenomenon can be explained as follows: 1) the molten alloy processes excellent fluidity and encroaches on the pore space when solidification, resulting in smaller practical porosity; 2) bigger pores store more un-melted powders and corresponding mass becomes larger, resulting in smaller measured porosity.

Moreover, the porosity measured by true densitometer is a little smaller than that measured by precision balance. This phenomenon can be explained as follows: pores inside samples can be classified into open pores and closed pores. The closed pores cannot be measured by true densitometer because it adopts Archimedes gas exhaust principle. While the closed pores can be measured by precision balance because it weights the solid part. As a result, true densitometer method leads to a smaller porosity compared to precision balance method.

Within the 3D-printed porous structures, porosities for D structure are quite different from theoretical ones regardless of measurement method applied. This phenomenon can be explained as follows: D structure possesses more sophisticated pore morphology than the others,

Table 3
Comparison of theoretical and practical porosity for TPMS structures.

Specimen structure	Theoretical porosity	Practical porosity					
		Precision balance method			True densitometer method		
		Mean value	Standard deviation	Relative error	Mean value	Standard deviation	Relative error
P	0.3	0.296	0.0155	1.3%	0.292	0.0113	2.7%
P	0.4	0.385	0.0127	3.8%	0.364	0.0216	9.0%
P	0.5	0.439	0.0143	12.2%	0.430	0.0156	14.0%
P	0.6	0.527	0.0074	12.2%	0.511	0.0178	14.8%
D	0.3	0.291	0.0112	3.0%	0.287	0.0077	4.3%
D	0.4	0.346	0.0124	13.5%	0.335	0.0133	16.3%
D	0.5	0.386	0.0161	22.8%	0.374	0.0198	25.2%
D	0.6	0.450	0.0214	25.0%	0.446	0.0108	25.7%
I-WP	0.3	0.288	0.0089	4.0%	0.283	0.0178	5.7%
I-WP	0.4	0.377	0.0122	5.8%	0.374	0.0133	6.5%
I-WP	0.5	0.461	0.0217	7.8%	0.455	0.0147	9.0%
I-WP	0.6	0.538	0.0086	10.3%	0.529	0.0046	11.8%

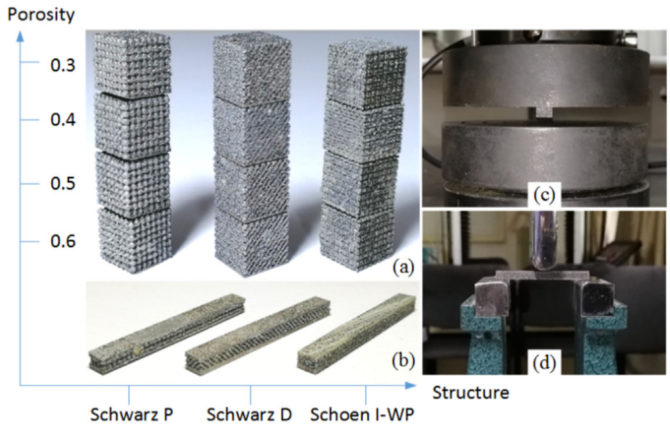

Fig. 6. Experimental specimens and equipment for mechanical evaluation: (a) compressive specimens; (b) bending specimens; (c) compressive equipment; (d) bending equipment.

making it more difficult to form, so the forming error is bigger than P and I-WP structures.

5. Mechanical evaluation of 3D-printed porous structure

5.1. Experimental method

Uniaxial compressive test is conducted to evaluate the compressive properties of 3D-printed wheel porous structures. Compressive specimens with shape of cube and size of 8 mm × 8 mm × 8 mm are designed and fabricated depending on SLM process, which are depicted in Fig. 6a. Every compressive specimen is composed of 512 TPMS units (1 mm × 1 mm × 1 mm). Three specimens are fabricated for every porosity and every TPMS structure. Universal mechanical test machine is utilized to perform the compressive test (Fig. 6c). The specimen is placed in the middle of lower platen. The lower platen is fixed and upper platen descends at the speed of 1 mm/min. Corresponding compressive stress-strain curves can be obtained and shown in Fig. 7a b and c. It shows a similar trend for three different structures, including approximate linear increase and fluctuating yield. Same as the normal porous material, the compressive properties of wheel structures will attenuate with the increase in porosity.

Three-point bending test is conducted to evaluate the flexural properties of 3D-printed wheel porous structures. Bending specimens with shape of cuboid and size of 36 mm × 4 mm × 3 mm are designed and fabricated based on SLM process as well, shown as Fig. 6b. Different from compressive specimen, bending specimen is designed as sandwich structure. In another word, TPMS structures (2 mm thickness) are sandwiched by upper and lower solid plates (0.5 mm thickness) to decrease stress concentration. Universal mechanical test machine with specific fixture (Fig. 6d) is applied for three-point bending test. The specimen is placed on the supporting rods whose span is adjusted to 30 mm. Then, pressing head descends at the speed of 0.5 mm/min to press the middle of specimen. Corresponding force-displacement curves can be obtained and shown as Fig. 7d e and f. It shows that the force will linearly increase with the increasing displacement until a sudden fall due to fracture failure. Similar to the compressive properties, the bending properties will attenuate with the increase in porosity.

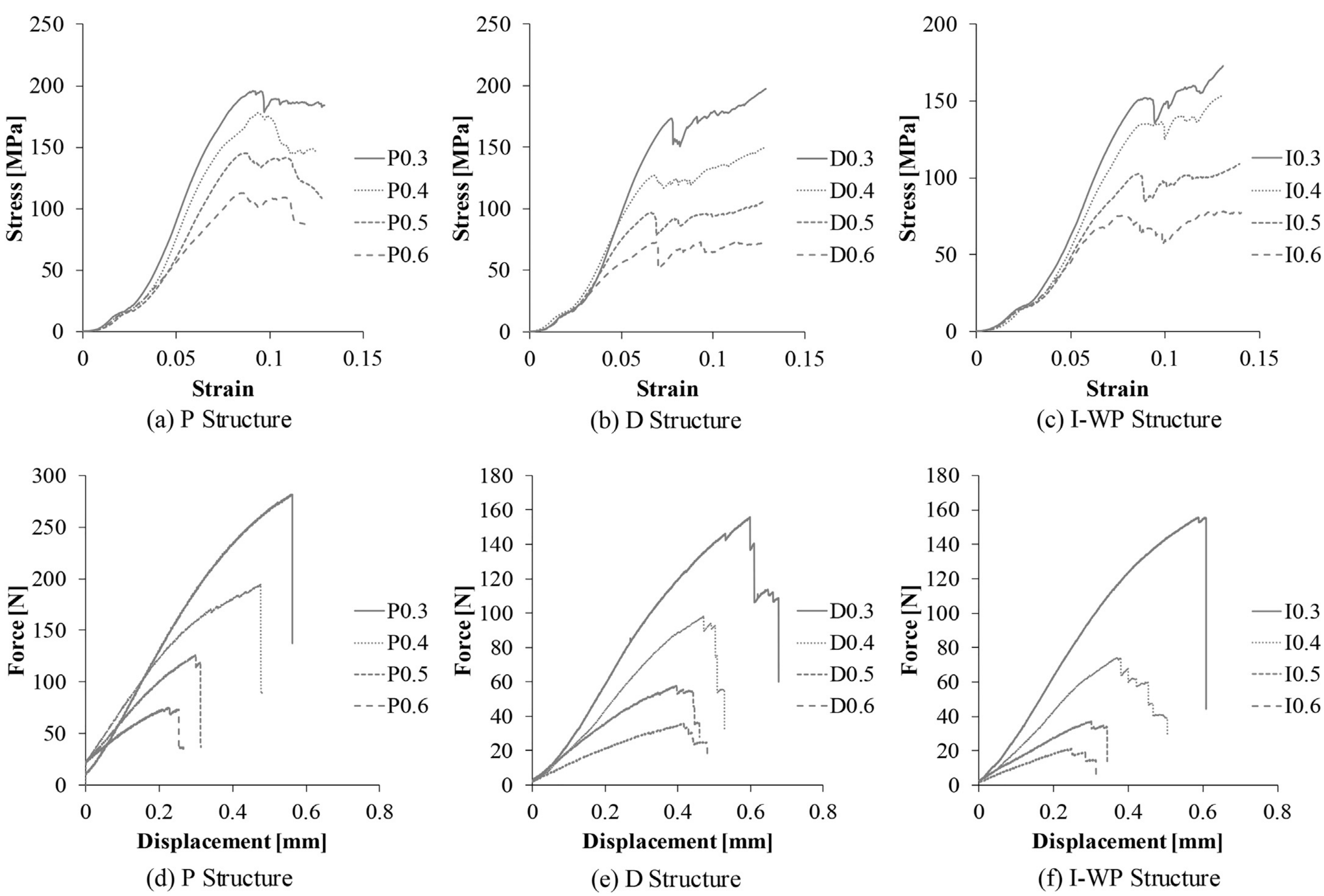

Fig. 7. Compressive stress-strain curves and bending force-displacement curves of 3D-printed wheel porous structures.

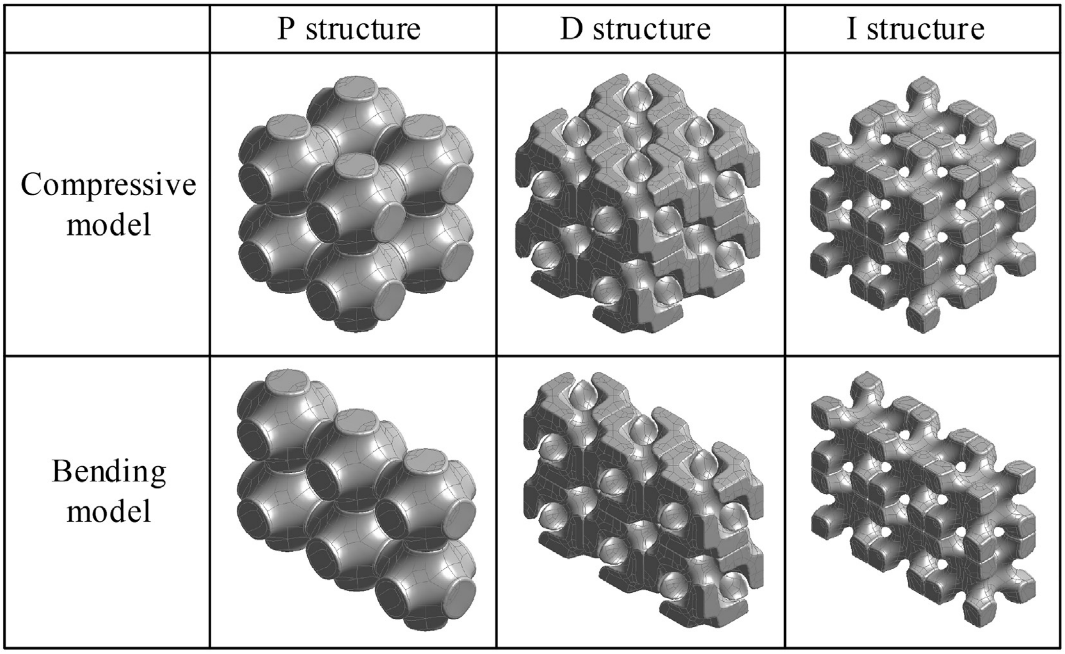

Fig. 8. 3D models of wheel porous structures used for finite element computing.

Table 4
Material definition in finite element modeling of wheel porous structure.

Property	Value	Unit
Density	2806	$kg \cdot m^{-3}$
Young's modulus	6.8	GPa
Poisson' ratio	0.31	–
Yield strength	410	MPa
Tangent modulus	2.0	GPa

5.2. Finite element method

Finite element method is adopted to further investigate and compare the mechanical properties of wheel porous structures in ANSYS platform. To simplify the 3D model and save the computing time, compressive model with $2 \times 2 \times 2$ TPMS units and bending model with $3 \times 1 \times 2$ TPMS units are utilized for the simulation and shown in Fig. 8.

Because the porous structures serve for the metal-bonded grinding wheel, the skeleton material is the composite composed of aluminum alloy and diamond abrasives, whose physical and mechanical properties are regarded as homogeneous and listed in Table 4. The density of composite is calculated from the density and volume fraction of its constituents. The mechanical properties are derived from the compressive test result of 3D-printed composite. Bilinear isotropic hardening model is adopted to describe the elastoplastic behavior of 3D-printed composite.

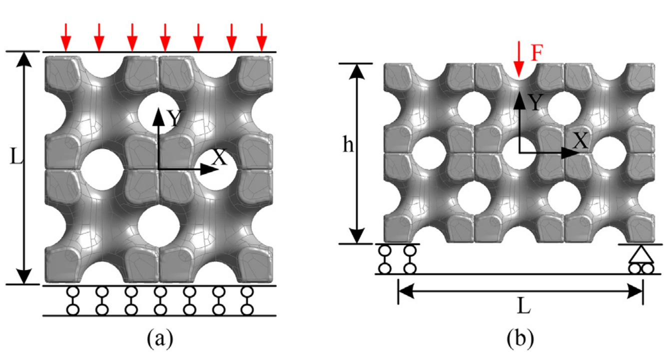

Fig. 9. Finite element model for compressive and bending simulation of wheel porous structure.

Boundary conditions are set for the finite element models and shown as Fig. 9. For the compressive model (Fig. 9a), the bottom surface is fixed with full constraints and the top surface is forced to descend along the negative direction of Y axis to the compressive strain of 10%. For the bending model (Fig. 9b), the left support constrains the translation in XYZ direction and does not constrain the rotational degree of freedom; the right support only carries out normal constraint and does not constrain other degrees of freedom. A forced displacement of 0.01 L along the negative direction of Y axis is applied at the middle position of the top surface, which makes the model undergo three-point bending deformation in the XY plane. Same as the experiment method, the compressive modulus (CM), yield strength (YS), bending modulus (BM) and bending strength (BS) can be obtained from finite element method.

5.3. Compressive properties analysis

According to ISO 13314 [16], the quasi-elastic gradient (QEG) is defined as the slope of linear deformation region at the beginning of the compressive stress-strain curves, which is used to characterize compressive stiffness. Another key parameter, called as first maximum compressive strength (FMCS), is defined as the first local maximum strength in the compressive stress-strain curves, which is used to characterize compressive strength. Therefore, based on the above definition, the two key mechanical indicators of Schwarz P structure, Schwarz D structure and Schoen I-WP structure with porosity ranging from 0.3 to 0.6 are calculated from the measured curves, and shown in Fig. 10a and b.

Fig. 10 shows the positive correlation exists for the compressive properties and relative density of every wheel porous structure. In another word, as the relative density increases, the QEG and FMCS increase as well. Besides, the Schwarz P structure possesses the best compressive properties in terms of QEG and FMCS. When the relative density is small, the Schoen I-WP structure behaves better than Schwarz D structure, and when the relative density is big, the contrary occurs.

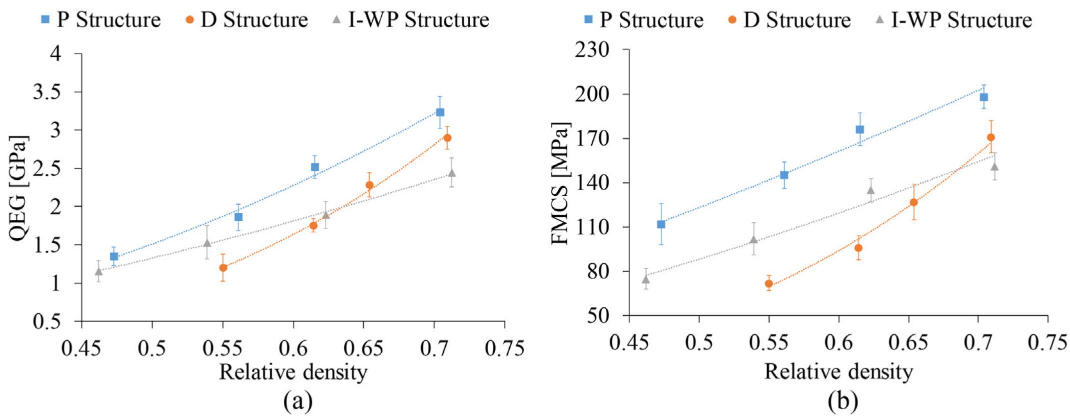

Fig. 10. QEG and FMCS of 3D-printed wheel porous structure.

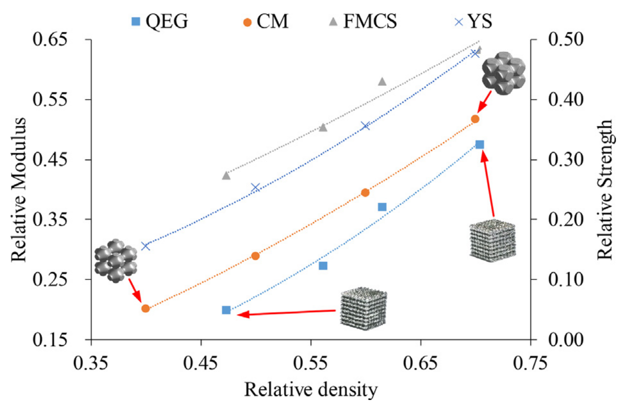

Fig. 11. Comparison of experimental and simulation values of compressive properties.

To eliminate the influence of skeleton material properties, the key mechanical indicators are normalized. The normalized mechanical indicators of porous structures are derived from the measured values divided by the corresponding mechanical indicators of the skeleton material. Taking the Schwarz P structure as an example, Fig. 11 shows the tendency of experimental values (QEG and FMCS) and simulation values (CM and YS) as the increment of relative density. It is found that the experimental values have similar tendency with the simulation values. To quantify the differences of compressive properties changing with relative density, the fitting equations are provided as follows:

$$\mathrm{QEG} = 1.0556\rho^{2.2492}$$

$$\mathrm{FMCS} = 0.8361\rho^{1.4763}$$

$$\mathrm{CM} = 0.9371\rho^{1.6839}$$

$$\mathrm{YS} = 0.9778\rho^{1.9843}$$

where ρ is the relative density of porous structures. It reveals that, with the same relative density, the values of QEG are slightly smaller than the values of CM. This is because the finite element method does not consider the material defect formed in practical manufacturing process. Besides, with the same relative density, the values of FMCS are slightly bigger than the values of YS. This is because FMCS represents the local maximum value of yield period while YS represents the beginning value of yield period.

1.1 Bending properties analysis

Similar as compressive properties, larger relative density contributes to better bending properties of wheel porous structures, which is in agreement with Gibson-Ashby theory [17]. The peak point of bending curves represents the initial crack generation and the beginning of fracture failure, from which the maximum deflection and fracture load can be derived. The bending modulus (BM) and bending strength (BS) are proportional to the linear gradient of curve and critical fracture force respectively when the span and size of sample are fixed [18].

The tendency of bending properties changing with relative density is displayed in Fig. 12. It reveals that the relative density significantly

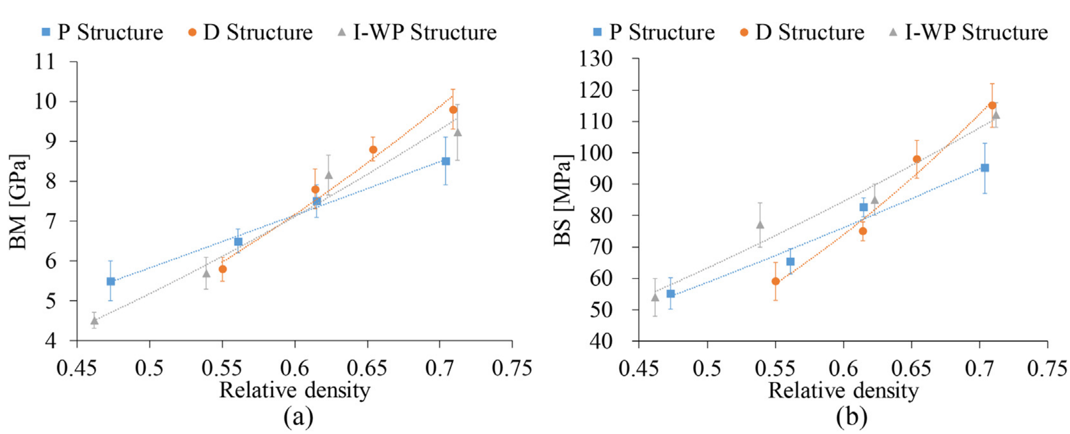

Fig. 12. BM and BS of 3D-printed wheel porous structure.

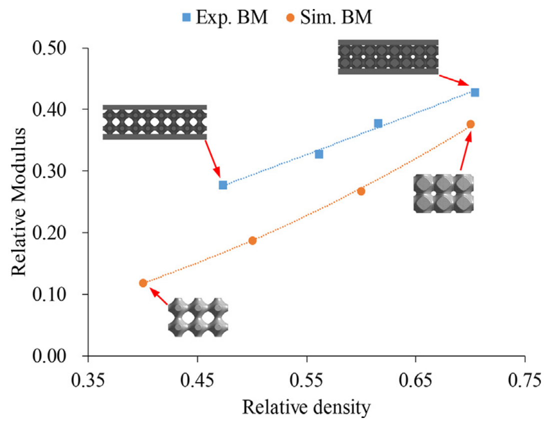

Fig. 13. Comparison of experimental and simulation values of three-point bending modulus.

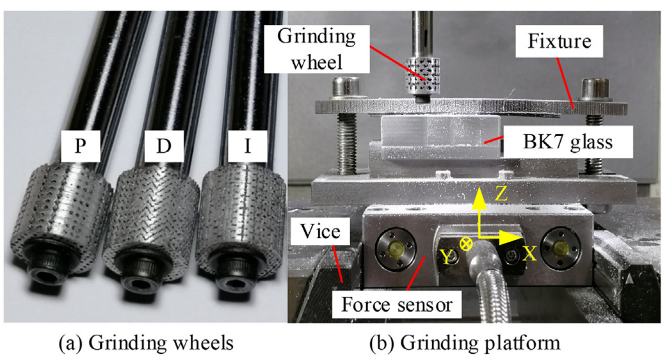

Fig. 14. Grinding wheels and grinding experiment platform.

Table 5
Grinding parameters.

Grinding wheel	Grinding mode	Grinding speed	Grinding depth	Feed rate	Grinding fluid
P, D, I, E	up-grinding	3 m/s	40 μm	20 mm/min	none

effects the bending properties of 3D-printed porous structure. The bending properties of Schwarz P structure, Schwarz D structure and Schoen I-WP structure are very closed to each other. Among the three TPMS structures, the increasing speed of bending properties of Schwarz D structure is the fastest as the increment of relative density. Therefore, when the relative density is small, the BM and BS of Schwarz D structure are at a low level; when the relative density is large, the BM and BS of Schwarz D structure are at a higher level.

The key indicators are normalized as well for bending properties. Taking the Schwarz P structure as an example, Fig. 13 shows the comparison of experimental and simulation values of three-point bending modulus. To quantify the differences of bending properties changing with relative density, the fitting equations are provided as follows:

$$\mathrm{Exp.BM} = 0.638\rho^{1.1183}$$

$$\mathrm{Sim.BM} = 0.7736\rho^{2.0451}$$

where ρ is the relative density of porous structures, Exp. BM represents the experimental value of BM, Sim. BM represents the simulation value of BM. It is revealed that the tendency is similar for the experimental and simulation values of BM, but the simulation BM is smaller than the experimental BM with the same relative density. This is because the structure of three-point bending specimens is different for experiment and simulation. In the bending experiment, the specimens are designed as sandwich structure to reduce the stress concentration, so the upper and lower plates increase the deformation resistance of the wheel porous structures and lead to a higher BM.

6. Grinding performance evaluation of 3D-printed wheels

To evaluate the grinding performance of 3D-printed grinding wheels, grinding experiment is conducted on BK7 glass. 3D-printed grinding wheels with size of 12mm × 12mm × 6mm are shown as Fig. 14a, which are compared to the electroplated grinding wheel with the same specification. P, D, I and E represent the grinding wheels with P structure, D structure, I-WP structure and electroplated wheel. Grinding experiment platform is depicted in Fig. 14b, including grinder, vice, fixture and force sensor. The grinding process parameters are listed in Table 5.

Fig. 15 shows that the grinding force and specific grinding energy of 3D-printed porous grinding wheels are both smaller than electroplated wheel by a large margin. Therefore, compared with electroplated grinding wheel, 3D-printed grinding wheel has higher energy efficiency, which can reduce energy consumption and heat production, and effectively improve grinding heat dissipation. The excellent performance of 3D-printed grinding wheel proves the effectiveness and advantages of the proposed novel design and fabrication method of grinding wheels.

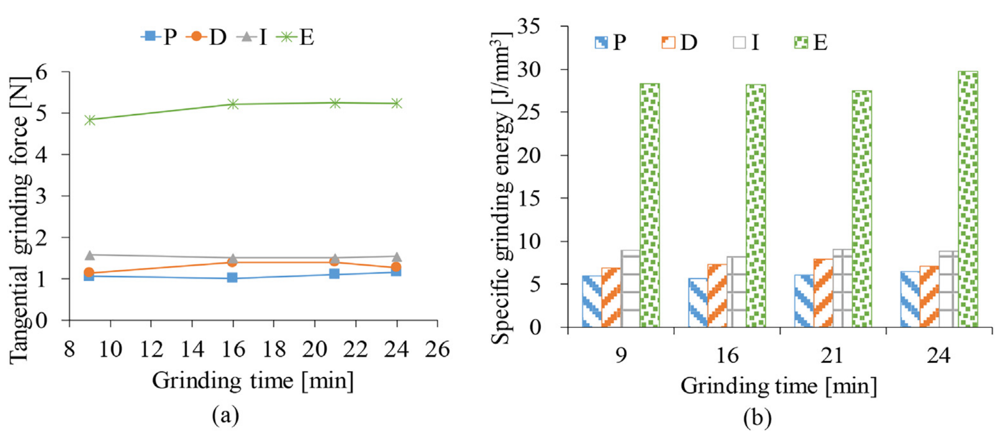

Fig. 15. Grinding force and specific grinding energy.

7. Conclusion

In this study, we propose one novel method to design and fabricate porous metal-bonded grinding wheel based on TPMS and 3D printing. The key findings are summarized as below:

(1) With the novel design method, grinding wheels with numerous interconnected pores and liquid channels can be realized. Furthermore, the porosity, relative density, equivalent aperture and specific surface area of wheel porous structure can be accurately expressed as the functions of offset value C. Based on the control equations, the geometric feature of wheel porous structure can be designed and regulated by changing the offset value C, therefore the pore morphology of wheel porous structure can be actively controlled.

(2) With the novel fabrication method, grinding wheels with well-embedded diamond abrasives and reliable bonding interface can be realized. Beside, wheel porous structure with different design parameters and well-formed pores can be realized. Due to the fluidity of molten metal and forming limits of SLM process, the practical porosity is lower than theoretical porosity for all the 3D-printed structures and the relative error becomes bigger as porosity increases.

(3) The mechanical properties (QEG, FMCS, BM and BS) of wheel porous structures will increase with the increment of relative density. The tendency is similar for the experimental values and simulation values of key mechanical indicators. Therefore, the mechanical properties of wheel porous structure can be controlled by changing design parameters.

(4) The grinding force and specific grinding energy of 3D-printed porous grinding wheels are both smaller than electroplated wheel by a large margin. The excellent performance of 3D-printed grinding wheel proves the effectiveness and advantages of the proposed method.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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